

External Adversarial Audit Report

Paper III - Linear-Error Clique Partitions of Split Graphs

Executive Verdict	Lean Block F	Primary Decision Point
PASS + OBS	CONFIRMED	No blocking defect found. Observation: Block B bibliography check is limited.

Lean cache	COMPLETED	lake exe cache get completed successfully; 8007 cached files decompressed.
Lean build	COMPLETED	lake build completed successfully: 8058 jobs.
Axiom gate	CLEAN	Layer E clean; Layer X uses AX1/AX2 exactly as expected.
Source scan	CLEAN	No sorry/admit/escape hatches; only AX1 and AX2 axiom declarations.
Finite scan	CONFIRMED	C-4/C-6 exact Fraction scan over p=3..80, q=1..2p, d=0..p.
Observation	PLAUSIBLE	Block B was source-page level, not full-paper bibliographic certification.

Archive SHA-256: f184f108d0f2e718a97518ddc9be42a6bdeac3f1c6d259ebdbefc605cbe8b14d EXTERNAL_AUDIT_RESULT_20260725_222145.zip

Per-Claim Decision Table

Claim	Math Audit	Lean	Evidence
Theorem 1.1 / E-9	Axiom-relative; no counterexample found	PASS	AX1+AX2 exactly in axiom report
Corollary 1.2	Axiom-relative; Lean signature checked	PASS	AX1+AX2 exactly in axiom report
Proposition 10.1	Internal audit rerun passed	PASS	Layer E clean
Theorem 3.1 / E-3.1	Exact F branch scan $p \leq 80$	PASS	Layer E clean
Theorem 4.2 / E-4.2	Exact margin scan $p \leq 80$	PASS	Layer E clean
Lemmas 5.1/5.2/6.1/7.1	Internal audit rerun passed	PASS	Layer E clean
Appendix B / D	Static/textual scan plus Lean gate	PASS	Layer E clean
AX1 / AX2	Limited bibliographic plausibility check	OBSERVATION	Declared axioms only; no extra axioms

Findings Register

ID	Block	Outcome	Severity	Finding / Coverage	Evidence
A-OBS-001	A	PLAUSIBLE	MINOR	case-insensitive label presence scan - all mandatory labels	blockA_faithfulness/results/blockA_results.txt
B-COV-001	B	PLAUSIBLE	MINOR	bibliographic source-page comparison - source pages only, not full PDF theorem audit	blockB_external_inputs/results/blockB_results.txt
C-CONF-001	C	CONFIRMED	NONE	exact Fraction scan - $p=3..80$, $q=1..2p$, $d=0..p$	blockC_counterexample_search/results/blockC_results.txt
D-CONF-001	D	CONFIRMED	NONE	exact Fraction identities - coefficients and threshold identities	blockD_algebra_rederivation/results/blockD_results.txt
E-CONF-001	E	CONFIRMED	NONE	re-run copied scripts after local dependency install - all <code>verify_*.py</code>	blockE_audit_the_audit/results/blockE_results.txt
F-CONF-001	F	CONFIRMED	NONE	lake cache/build/gate plus source scan - frozen bundled snapshot executed from short local path	blockF_lean_formalization/results/blockF_summary.txt

Lean Gate Detail

Block F completed locally from C:\Users\jtrav\Documents\Codex\leanrun_erdos81. The table below extracts the decision-relevant axiom dependencies from gate.lean.

Declaration	Axiom Dependencies	Decision
PaperIII.Theorem_1_1	propext, Classical.choice, PaperIII.AX1, PaperIII.AX2, Quot.sound	PASS
PaperIII.Corollary_1_2	propext, Classical.choice, PaperIII.AX1, PaperIII.AX2, Quot.sound	PASS
PaperIII.E_4_3	propext, Classical.choice, PaperIII.AX1, Quot.sound	PASS
PaperIII.E_8	propext, Classical.choice, PaperIII.AX1, PaperIII.AX2, Quot.sound	PASS
PaperIII.Prop_10_1_low	propext, Classical.choice, Quot.sound	PASS
PaperIII.Prop_10_1_mid	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_3_1	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_4_1	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_4_2	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_5_1	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_5_2	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_6_1	propext, Classical.choice, Quot.sound	PASS
PaperIII.E_7_1	propext, Classical.choice, Quot.sound	PASS
PaperIII.pathCorrection_odd_iff	propext, Classical.choice, Quot.sound	PASS
PaperIII.AppendixD.konig_edge_coloring	propext, Classical.choice, Quot.sound	PASS
PaperIII.AppendixD.galvin_max_degree	propext, Classical.choice, Quot.sound	PASS
PaperIII.clique_divisible_correction	propext, Classical.choice, Quot.sound	PASS
PaperIII.AX1	propext, Classical.choice, PaperIII.AX1, Quot.sound	PASS
PaperIII.AX2	propext, Classical.choice, PaperIII.AX2, Quot.sound	PASS

Coverage And Caveats

Status	Scope
COMPLETED	Block F Lean build/gate/source scan completed and is the main improvement over the prior inconclusive report.
COMPLETED	Block E internal audit scripts reran successfully after local dependency installation.
COMPLETED	Block C exact finite formula scan found no violations over p=3..80, q=1..2p, d=0..p.
PLAUSIBLE	Block B remains limited to source-page bibliography checks; this is why the report says PASS_WITH_OBSERVATIONS rather than an unqualified PASS.
NOT CHANGED	This PDF changes only formatting and readability. It does not change claims, logs, findings, or evidence files.

Key log anchors

Build log contains: Build completed successfully (8058 jobs).

Block F summary contains: cache_get_ok=True, lake_build_ok=True, gate_ok=True, static_scan_ok=True.